a Observations



SWOT: WSE · width · slope

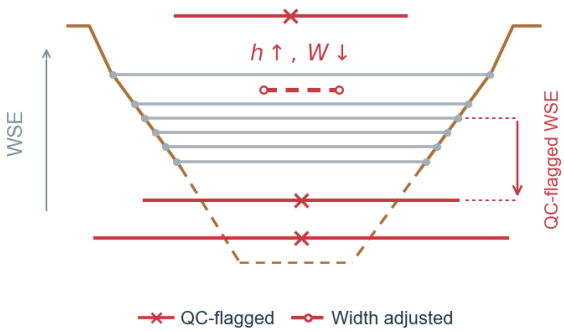


Gauge discharge Q

Photo: Tim Merrick / USGS

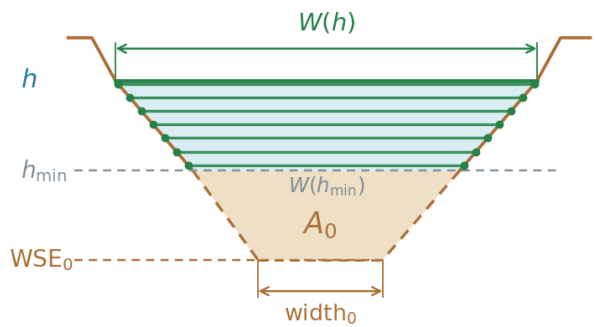
b Width correction and hydraulic parameterization

Raw observations



Schematic cross-sections

Screened + corrected



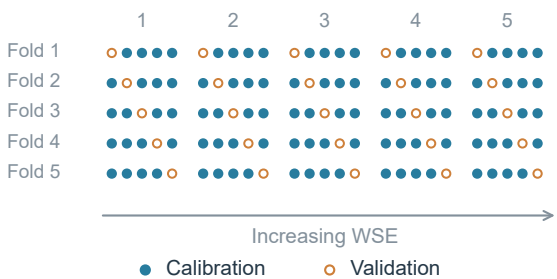
$$A(h) = A_0 + \beta \int_{h_{\min}}^{h} W(z)\,\mathrm{d}z \quad (h \geq h_{\min})$$

$$Q = \frac{1}{n} A(h)^{5/3} W(h)^{-2/3} S^{1/2}$$

c Calibration and validation

5 WSE strata · 5 validation folds

About 80% calibration / 20% validation



Calibration WSE range



Fixed W(h) and hydraulic parameters

Each observation is validated once

Across-gauge NSE and RMSE: median ± IQR/2 over five folds. Gauge-level scores use all out-of-fold predictions.